

Functional & Performance Testing Template

Model Performance Test

Date	3 July 2026
Project Name	Smart Lender - Loan Eligibility Prediction System
Maximum Marks	

Test Scenarios & Results

Test Case ID	Scenario (What to test)	Test Steps (How to test)	Expected Result	Actual Result	Pass/Fail
FT-01	Categorical field validation (Gender, Married, Education, Self_Employed, Property_Area)	Select each dropdown option and submit the form	All valid dropdown selections accepted; prediction generated without error	All valid selections accepted and processed correctly	Pass
FT-02	Numeric input validation (Applicant Income, Loan Amount, Loan Term)	Enter numbers within typical range and an unusually high value (e.g. Loan Amount = 10000)	Accepts valid values; handles out-of-range values without crashing	Valid values processed correctly; high value also produced a valid prediction	Pass
FT-03	Prediction generation (core feature)	Fill in complete applicant details and click 'Assess Application'	Correct Approved/Rejected result generated based on input	Correct result generated with confidence score displayed	Pass
FT-04	Model/Backend connection check	Start Flask app and verify the model loads and responds to a form submission	App starts successfully and model responds with a prediction	App started on http://127.0.0.1:5000 and responded correctly	Pass
PT-01	Response Time Test	Use a timer to measure time from clicking 'Assess Application' to result display	Should be under 3 seconds	Prediction returned in under 1 second (model loaded once at startup)	Pass
PT-02	Repeated Request Speed Test	Submit multiple predictions in quick succession	Application should not slow down or crash	Each request processed independently without slowdown	Pass
PT-03	Data Load Test	Run the training script on the full dataset (491 rows) and check processing	Should process smoothly without crashing	All 4 models trained successfully on the full dataset without errors	Pass

Summary

All functional test cases (FT-01 to FT-04) and performance test cases (PT-01 to PT-03) passed successfully. The application correctly validates inputs, generates accurate predictions, and responds within acceptable time limits (sub-second per request) since the trained model is loaded once at application startup rather than reloaded per request.